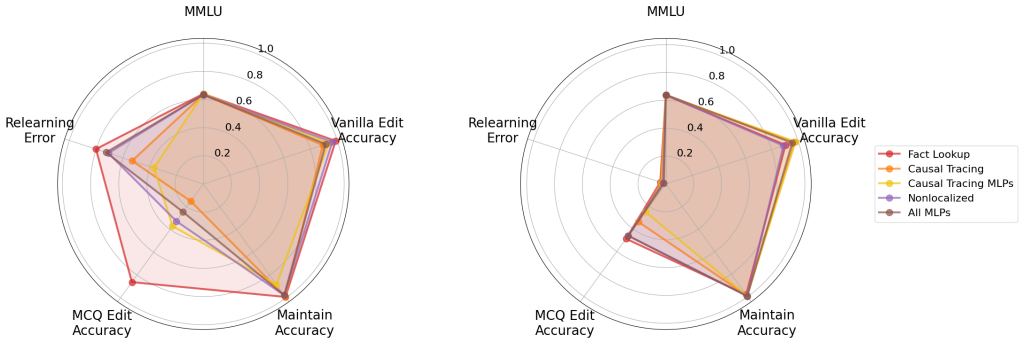
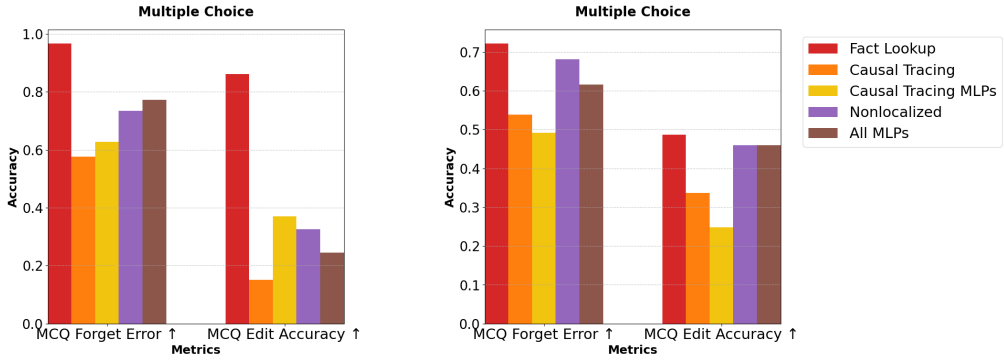


(a) Sports-Athlete-Editing (b) Full-Sports-Editing

Figure 2: Spider plots illustrating the advantages of FLU for editing Sports across adversarial prompting and relearning evaluations, averaged over all three model types. **(Left)** The Sports-Athlete-Editing plot shows that FLU localization leads to editing that is the most robust against MCQ prompting and relearning. **(Right)** The plot shows that most localizations perform approximately equivalently in the Full-Sports-Editing task, with FLU localization slightly better for MCQ.



(a) Sports-Athlete-Editing (b) Full-Sports-Editing

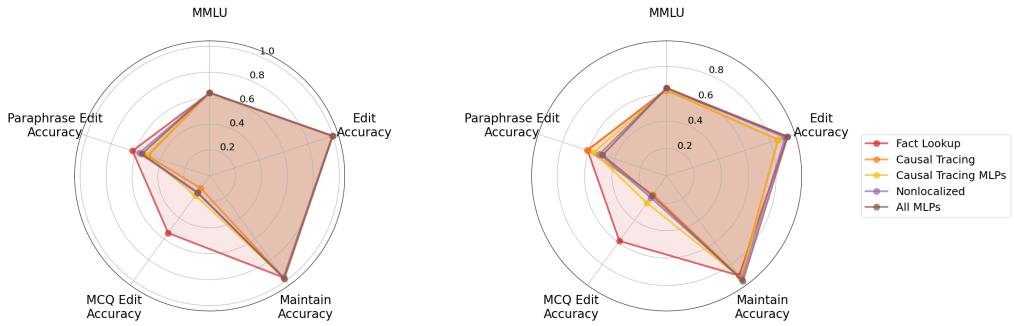
Figure 3: Bar charts showing results of MCQ evaluations, reporting both the forget error and edit accuracy when prompted with MCQ, averaged over three model types. For both **(a)** Sports-Athlete-Editing and **(b)** Full-Sports-Editing, FLU localization answers with the original answer the least (MCQ Forget Error) and answers with the edited answer most accurately (MCQ Edit Accuracy).

Sports Results Our analysis reveals that editing employing FLU localization exhibits superior performance in forgetting the original information and adopting the edited information, across different prompt formats. As explained in Section 3.1, better editing should result in higher *MCQ Forget Error*, and higher *MCQ Edit Accuracy*. Figure 3 shows that FLU localized models are the best on both fronts. The difference is especially significant in Figure 3a, where only FLU model edits generalize meaningfully to MCQ, exceeding other localization methods by more than 40% in MCQ Edit Accuracy.

A comprehensive comparison of all localization methods with multiple-choice prompting is available in Appendix A.7.2, further supporting our findings.

3.1.2 COUNTERFACT TASKS

CounterFact Prompting For CounterFact, we create an MCQ evaluation with four choices for every question, randomly ordering the true answer, the injected false answer, and two other question-specific LLM-generated incorrect answers. We also consider the original robustness and side effect evaluations from the Meng et al. (2023) dataset: the Paraphrase and Neighborhood facts accuracies. Answers of edited facts are meant to generalize to the Paraphrase evaluation, which phrases the fact in a different but equivalent way, so we report the *Paraphrase Edit Accuracy* (stronger methods



(a) CounterFact-Editing (b) Sequential-CounterFact-Editing

Figure 4: Spider plots illustrating the advantages of FLU for editing CounterFact across prompting evaluations, averaged over all three model types. **(Left)** The CounterFact-Editing plot shows that FLU localization leads to editing that is the most robust against MCQ prompting and Paraphrasing. **(Right)** The Sequential-CounterFact-Editing plot shows that FLU localization is the most robust against MCQ prompting.

should increase Paraphrase edit accuracy). Editing should not generalize to the Neighborhood evaluation, which presents similar but unrelated facts to the forget set, so we report the *Neighborhood Edit Error* which is lower if models incorrectly report the edited answer in these unrelated facts.

We again also use an MCQ evaluation, where the choices consist of the true answer, the injected false answer, and two other question-specific LLM-generated incorrect answers. We note that Paraphrase and Neighborhood evaluations ask for the answer in the same original format, so they are more in-distribution than MCQ. We average accuracies over all models, and for CounterFact-Editing we average over editing both 16 and 64 facts.

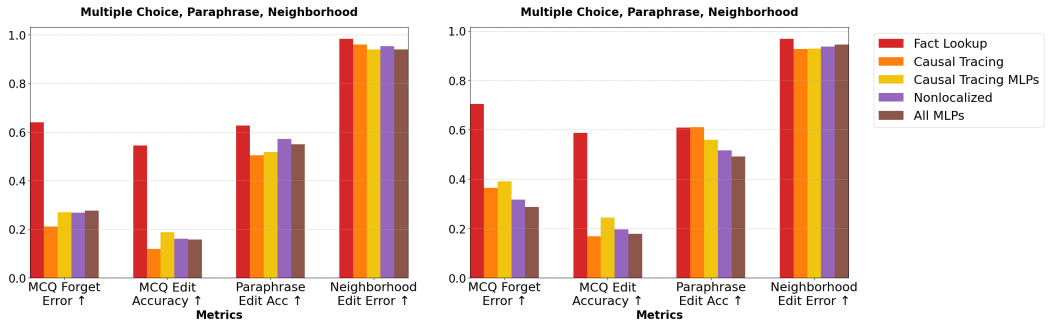
CounterFact Results Figure 5 illustrates the robustness of FLU editing in the MCQ and Paraphrase evaluations. Edited models using FLU localization answer less frequently with the original, incorrect information (*MCQ Forget Error*) and more frequently provide the new, edited answer. Furthermore, the *Neighborhood Edit Error* highlights that other localization methods exhibit slightly more pronounced side effects, inadvertently editing unintended, semantically similar facts.

Interestingly, sequential editing displays marginally greater robustness than nonsequential editing in MCQ when comparing between CounterFact-Editing and Sequential-CounterFact-Editing bars. This observation supports an approach that editing large sets of facts can be made more effective by partitioning the set and applying edits sequentially. We present results comparing all localizations across each prompt robustness evaluation in Appendix A.7.2, with the largely consistent conclusions.

3.2 Adversarial Relearning Evaluation

We measure the ability of our models to withstand adversarial relearning, both to address the scenario in which adversaries may have fine-tuning access and as a measure for the quality of editing. We replicate the methodology of Deeb & Roger (2024), splitting our forget sets in two independent halves, retraining with half of the ground truth labels, and evaluating on the other half. This methodology aims to discern whether the editing technique successfully removed the underlying factual association or merely obfuscated its direct retrieval while leaving it potentially susceptible to recovery when doing partial retraining.

We retrain with a rank-512 LoRA across all linear modules, with details available in Appendix A.7.3. In this section, we focus on the Sports-Athlete-Editing task, as in the other tasks it was either too easy to relearn (Full-Sports-Editing) or too hard to relearn any performance (all CounterFact tasks) across all localizations. Relearning isn't a valid evaluation for Full-Sports-Editing because the facts are not independent, and models should reasonably generalize from relearning on half the basketball athletes to answering with basketball on the other half of the basketball athletes. We show results from Counterfact and from all localizations in Appendix A.7.3.



(a) CounterFact-Editing (b) Sequential-CounterFact-Editing

Figure 5: Bar charts showing results of MCQ, Paraphrase, and Neighborhood prompt evaluations, averaged over all three model types. For both **(a)** CounterFact-Editing and **(b)** Sequential-CounterFact-Editing, FLU localization has the most robust edit accuracy measured by MCQ and Paraphrase. FLU localization editing also does not incorrectly generalize to Neighborhood prompts. Sequential editing is slightly more robust than nonsequential editing in MCQ when comparing between CounterFact-Editing and Sequential-CounterFact-Editing bars.

Sports Results Our adversarial relearning experiments, as depicted in Figure 6, reveal that retraining on a subset of the original "forgotten" data can recover a significant portion, as much as 63%, of the supposedly forgotten information when using OT methods like Causal Tracing and Causal Tracing MLPs. This suggests that these methods may simply mask direct retrieval of this information, leaving the model susceptible to this information recovery through retraining. In contrast, FLU localization exhibits greater resilience to such adversarial relearning, with only about 20% of the forgotten information recovered. This indicates that FLU localization may be more effective in targeting and removing the underlying knowledge, making it harder to recover through retraining.

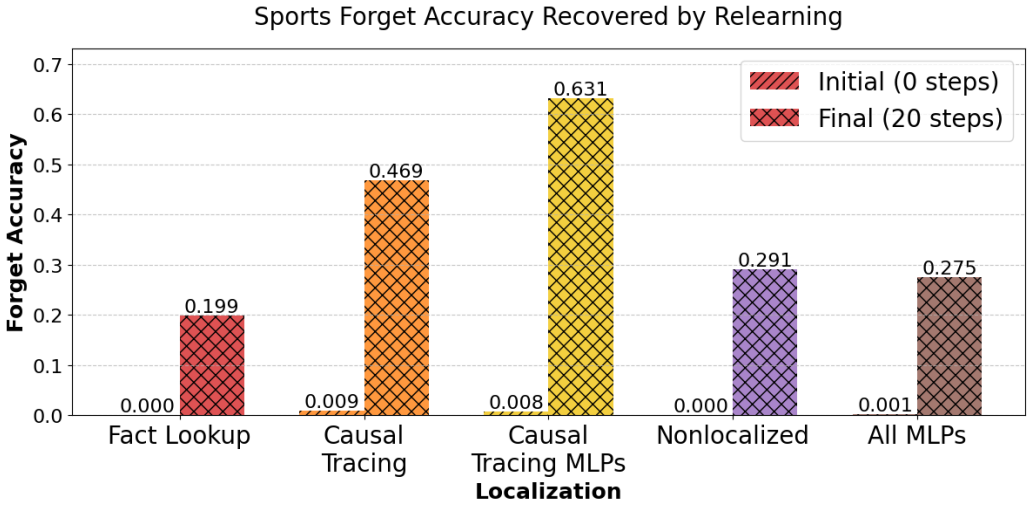


Figure 6: Relearning recovers the least accuracy on the forget set using FLU localizations. Relearning recovers significant accuracy on the original forget set in OT localizations (Causal Tracing and Causal Tracing MLPs).